

i is reg-index in the code

push local i // addr = LCL + i @ LCL D = M @ i D = D + A // *SP = *addr A = D D = M @ SP A = M M = D // SP++ → @SP M = M + 1	@ addr A = M M = D Similarly for argument, this, that push constant i // *SP = i @ i D = A @ SP A = M M = D // SP++ @ SP M = M + 1	pop static i // SP-- @ SP M = M - 1 // foo.i = *SP A = M D = M @ foo.i M = D push temp i // addr = 5 + i @ 5 D = A @ i D = D + A // *SP = *addr A = D D = M @ SP A = M M = D // SP++ @ SP M = M + 1
Similarly for argument, this, that		pop temp i // addr = 5 + i @ 5 D = A @ i D = D + A
pop local i // addr = LCL + i @ LCL D = M @ i D = D + A @ addr M = D // SP-- M = M - 1 // *addr = *SP A = M D = M	push static i // *SP = foo.i @ foo.i D = M @ SP A = M M = D // SP++ @ SP M = M + 1	@ SP A = M M = D // SP++ @ SP M = M + 1 pop temp i // addr = 5 + i @ 5 D = A @ i D = D + A

@ addr	add	$A = A - 1$
$M = D$	@ SP	$M = -1$
// SP--	$A = M - 1$	@ EQ
@ SP	$A = A - 1$	D; JEQ
$M = M - 1$	$D = M$	@ SP
// *addr = *SP	$A = A + 1$	$A = M - 1$
$A = M$	$D = D + M$	$A = A - 1$
$D = M$	$A = A - 1$	$M = 0$
@ addr	$M = D$	(EQ)
$A = M$	@ SP	@ SP
$M = D$	$M = M - 1$	$M = M - 1$

push pointer 0/1
 // *SP = THIS/THAT
 @ THIS/THAT
 $D = M$
 @ SP
 $A = M$
 $M = D$
 // SP++
 @ SP
 $M = M + 1$

Similarly for sub
 and, or. just
 replace '+' in
 $D = D + M$

not
 @ SP
 $A = M - 1$
 $M = !M$

Similarly for gt and
 lt, just replace
 JEQ with JGT or
 JLT and give a
 generic name to the
 label.

pop pointer 0/1
 // SP--
 $M = M - 1$
 // THIS/THAT = *SP
 $A = M$
 $D = M$
 @ THIS/THAT
 $M = D$

Similarly for
 neg

eg
 @ SP
 $A = M - 1$
 $A = A - 1$
 $D = M$
 $A = A + 1$
 $D = D - M$